Karnataka ReddyJana Sangha®

VEMANA INSTITUTE OF TECHNOLOGY

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DEPARTMENT OF CHEMISTRY

CHEMISTRY LABORATORY MANUAL

Applied Chemistry for Emerging Electronics and futuristic Devices

(*EEE Stream*, Couse code: 1BCHEE202)

(Second Semester B.E)

Student Name : …….

USN : …….

Branch/ Section : ..... Year: 2025-26

APPLIED CHEMISTRY LABORATORY MANUAL

(As per Choice Based Credit System (CBCS) Scheme)

SEMESTER –II

Course Code	1BCHEE202
CIE Marks (IA Marks)	15+10
Number of Hours/ Week	02

List of Experiments

Sl No	Experiment Name
1	Estimation of total hardness of water by EDTA method.
2	Determination of chemical oxygen demand (COD) of industrial effluent sample.
3	Estimation of iron in TMT bar by diphenyl amine indicator method.
4	Determination of total alkalinity of given water sample.
5	Determination of Hydrogen Peroxide in PCB Cleaner (Permanganate method)
6	Estimation of acid mixture by conductometric sensor (Conductometry)
7	Estimation of iron in rust sample by Potentiometric sensor (Potentiometry)
8	Determination of pKa of vinegar using pH sensor (Glass electrode)
9	Estimation of copper present in e-waste by optical sensor (Colorimetry).
10	Determination of Viscosity coefficient of conductive links
11	Smartphone-Based colorimetric estimation of total phenolic content in coffee products.
12	Green synthesis of copper nanoparticles for conductive ink applications.

Department of Chemistry

GENERAL INSTRUCTIONS

- Keep your working table clean and tidy and your apparatus scrupulously clean.
- Keep gas taps and water taps closed when not in use.
- Use either spirit lamps or candle to light burner. Do not use paper torches for the purpose.
- Do not throw any waste paper or litmus paper into the sink. Throw them into the dustbin provided.
- Do not pour concentrated acid into the sink. If it has to be poured, flush using water liberally.
- Record your observations in an observation book as and when you observe and notafter completion and keep it away from the reagents and sink.
- Use the correct quantity of reagents. Do not use wet hand during weighing.

VOLUMETRIC ANALYSIS

BURETTE

- Wash the burette with tap water and then rinse with a small quantity of solution to be taken in the burette anddiscard the solution into the sink.
- Fill the burette with a solution using a small funnel, which is already washed with water.
- Record the initial reading by seeing the level of the burette solution at your eye level to avoid the error due to parallax.

PIPETTE

- Wash the pipette with tap water and then rinse with a small quantity of the solution to be pipette and discard the solution into the sink.
- Pipette out exactly 25 cm^3 of the solution and do not suck beyond the mark point.
- During transferring of solution to the clean conical flask, touch the tip of the pipette to the bottom of the conical flask gently after complete transfer of solution.
- Do not blow out the last drops of the liquid from the pipette.

CONICAL FLASK

- Wash the conical flak with tap water and then rinse with distilled water.
- Do not rinse the conical flask with solution to be taken in the conical flask.
- During the titration, place the conical flak upon a glazed white tile beneath the burette.
- During the addition of the solution from the burette, conical flask must be constantly rotated with one hand, while other hand has to control the knob of the burette.

ESTIMATION OF TOTAL HARDNESS OF WATER BY EDTA METHOD

OBSERVATION & CALCULATION:

Weight of weighing bottle + EDTA salt (W_1) = ... g
Weight of empty weighing bottle (W_2) = ...g
Weight of EDTA salt Transferred (W_1- W_2) = ... g

$$\text{Molarity of EDTA Solution} = \frac{\text{Weight of EDTA } (W_1\text{-}W_2)\text{x } 4}{\text{Molecular weight of EDTA } (372.24)}$$

$$= \frac{\text{x4}}{372.24}$$

$$= _____ \text{M (z)}$$

Burette readings	Trial I	Trial II	Trial III
Final burette reading (cm^3)			
Initial burette reading (cm^3)	0.0		
Volume of EDTA solution Consumed (cm^3)			

Volume of EDTA consumed = ...cm^3 (x)

1000 cm^3 of 1M EDTA = 100 g (Molecular mass of $CaCO_3$ = 100)

……x…cm^3 of …z….M EDTA = $\frac{\text{…x……..cm}^3 \text{ x …z…… x 100}}{1000}$ g of $CaCO_3$

25 cm^3 of the water sample contains …… g of $CaCO_3$

Total hardness of the given water sample = $\frac{\text{g x } 10^6}{25}$ ppm of $CaCO_3$

= .....ppm of $CaCO_3$

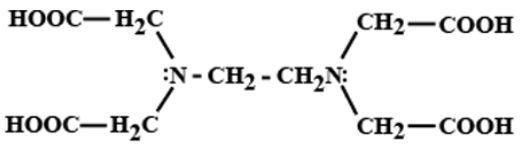
Result: Total hardness of the given water sample = ----------ppm $CaCO_3$

Date			
Experiment No.	1		

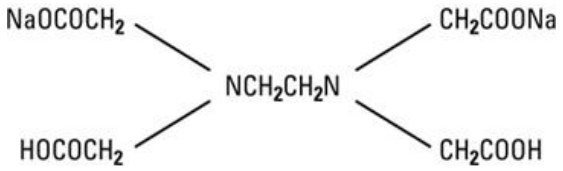
ESTIMATION OF TOTAL HARDNESS OF WATER BY EDTA METHOD

Aim: To estimate the total hardness of given water sample by EDTA method

Principle:
Hardness of water is due to the presence of calcium and magnesium salts in water. Ethylene diamine tetra acetic acid (EDTA) forms complex with a large number of cat ions including Ca^{2+} and Mg^{2+} ions. So it is possible to determine the total hardness of water using EDTA reagents.



stucture of EDTA



The disodium salt of EDTA molecule (H_2Y) has two easily replaceable hydrogen atoms and the after ionization it is represented as H_2Y^{2-}. The later forms complexes with metal ions as follows.

$$M^{2+} + H_2Y^{2-} \longrightarrow MY^{2-} + 2H^+ \cdots\cdots\cdots\cdots (1)$$

Where M^{2+} is Ca^{2+} and Mg^{2+} in water. Reaction (1) can be carried out quantitatively at a P^H of 10 using Eriochrome black-T indicator. Since the reaction involves the liberation of H^+ ions and hence buffer mixture need to be used to maintain a pH of 10. The buffer mixture used in the titration is $NH_3 – NH_4Cl$. Since EDTA (free acid) is sparingly soluble, its disodium salt, Na_2H_2Y used for preparing the reagent. The hardness of water is usually expressed in terms of ppm (Partsper million) of $CaCO_3$.

Procedure:
<u>Part A- Preparation of a Standard EDTA Solution</u> – Weigh accurately the given EDTA crystals using an electronic weighing balance. Note the weight, transfer the crystals carefully into a funnel placed over a 250 cm^3 volumetric flask and note down the empty weight of the weighing bottle. Add 2 cm^3 of ammonia and pour ion exchange water through the funnel allowing all the crystals to run down into the flask. Wash the funnel with ion exchange water and remove the funnel. Dissolve the crystals by swirling the flask gently. Add some more waterif needed. Dilute the solution up to the mark with ion exchange water, stopper the flask and mix the solution thoroughly by inverting the flask several times so that a homogeneous solutionresults. Calculate the molarity of EDTA.

<u>Part B - Determination of total hardness of a sample of water</u> - Pipette out 25 cm^3 of the given water sample into a clean conical flask. Add 2 cm^3 of $NH_3 – NH_4Cl$ buffer and 2 drops of Eriochrome black – T indicator. Titrate against standard EDTA solution till the colour of the solution changes from wine red to clear blue. Repeat the titration for concurrent values.

Result: Total hardness of the given water sample = ---------- ppm $CaCO_3$

DETERMINATION OF CHEMICAL OXYGEN DEMAND (COD) OF INDUSTRIAL EFFLUENT SAMPLE

OBSERVATION & CALCULATION:

Weight of weighing bottle + FAS = W_1 =_____g

Weight of empty weighing bottle = W_2 =_____g

Weight of FAS transferred (W) = W_1-W_2 =_____g

$$\text{Normality of Mohr's salt} = \frac{\text{Weight of FAS } (W_1\text{-}W_2) \times 4}{\text{Equivalent weight of Mohr's salt } (392.14)}$$

$$= \frac{\quad \times 4}{392.14}$$

= .....N (z)

Burette readings	Trial I	Trial II	Trial III
Final burette reading (cm^3)			
Initial burette reading (cm^3)			
Volume of FAS (cm^3)			

Volume of FAS solution consumed =..... cm^3 (Y)

The blank titrate value = ...cm^3 (X)

1000 cm^3 of 1 N Mohr's salt solution = 1g equivalent of oxygen, i.e, 8 grams of oxygen

(x-y) cm^3 of ------- N Mohr's salt solution = $\frac{8\,(X\text{-}Y)\,x}{25}$.....$\frac{N \times 1000}{}$

= .....

Chemical Oxygen Demand of the given waste water sample =…….. mg of oxygen / dm^3

Result:

Chemical Oxygen Demand of the given waste water sample =.....mg of oxygen/dm^3

Date			
Experiment No.	2		

DETERMINATION OF CHEMICAL OXYGEN DEMAND (COD) OF INDUSTRIAL EFFLUENT SAMPLE

Aim: To determine the Chemical Oxygen Demand (COD) of industrial efflunet sample

Principle:

Chemical Oxygen Demand is a measure of the oxygen equivalent of that portion of organic and inorganic materials that is oxidized by a strong oxidizing agent. This parameter is important for the study of industrial waste water and control of waste treatment plants.

Waste water contains organic and inorganic impurities which include straight chain aliphatic compounds, aromatic hydrocarbons, alcohols, acids, pyridine and other oxidisable materials. Straight chain compounds, acetic acid etc. are oxidized more effectively when silver sulphate is added as a catalyst. But silver sulphate reacts with chlorides in the waste water to form a precipitate of silver chloride, which are oxidized only partially by this procedure. This difficulty is overcome by adding mercuric sulphate to the sample.

Procedure:

PART- A: Preparation of standard Mohr's salt (FAS) solution - Weigh out given Mohr's salt (FAS) accurately and transfer into a 250 cm^3 volumetric flask through a glass funnel. Add 2 test tube of dilute H_2SO_4 and little quantity of ion exchange water. Remove the funnel, dissolve the salt and dilute up to the mark with ion exchange water and mix well. Calculate the Normality of Mohr's salt (FAS).

PART -B: Determination of COD (Back Titration) - Pipette out 25 cm^3 of the waste water sample into a 250 cm^3 conical flask. Add 10 cm^3 standard potassium dichromate solution followed by 1 test tube of 1:1 sulphuric acid (Containing silver sulphate) with constant shaking of the contents of the flask. Add 2-3 drops of ferroin indicator and titrate against the Mohr's salt solution until the solution turns bluish green to reddish brown (Brick Red).

PART – C: Blank Titration - Perform the blank titration by titrating 10 cm^3 of $K_2Cr_2O_7$ solution against Mohr's salt solution as above but without using waste water sample. Calculate the COD of waste water sample by taking difference between the blank and back titration values.

Result: Chemical Oxygen Demand of the given waste water sample = mg of oxygen/dm^3

ESTIMATION OF IRON IN TMT BAR BY DIPHEYNYL AMINE INDICATOR METHOD

OBSERVATION & CALCULATION

Weight of bottle + $K_2Cr_2O_7$ W_1 =_____g

Weight of empty bottle W_2 = _____g

Weight of $K_2Cr_2O_7$ crystals transferred ($W_1 - W_2$) =_____g

Equivalent weight of $K_2Cr_2O_7$ = 49

$$\text{Normality of } K_2Cr_2O_7 \text{ solution} = \frac{\text{Mass of } K_2Cr_2O_7 \times 4}{\text{Eq. weight of } K_2Cr_2O_7 \ (49)}$$

$$= \frac{g \times 4}{49}$$

=_____N (y)

Burette readings	Trial I	Trial II	Trial III
Final burette reading (cm^3)			
Initial burette reading (cm^3)			
Volume of Potassium Dichromate (cm^3)			

The weight of iron ore in 250 cm^3 of the given solution =_____g (W) will be given

The volume of $K_2Cr_2O_7$ =_____cm^3 (x)

1000 cm^3 of 1N $K_2Cr_2O_7$ = 1 milli equivalent of Fe

1000 cm^3 of 1N $K_2Cr_2O_7$ = 55.85 g of Fe

…….cm^3 of .....N $K_2Cr_2O_7$ $= \frac{55.85 \times cm^3 \times N}{1000}$

= .....g of Fe (a)

250cm3 of the solution contains 10x (a) g of Fe =10x (a) g =_____g (b)

Percentage of Iron in the given 250 cm^3 of solution $= \frac{b \times 100}{W}$

= _____ %

Result: Amount of iron in given TMT bar =_____%

Date			
Experiment No.		3	

ESTIMATION OF IRON IN TMT BAR BY DIPHEYNYL AMINE INDICATOR METHOD

Aim: To estimate the amount of iron in TMT bar by diphenyl amine indicator method

Principle:

Procedure:

<u>Part A: Preparation of standard potassium dichromate solution</u> - Weigh out accurately given $K_2Cr_2O_7$ crystals and transfer to 250 cm^3 volumetric flask. Dissolve in ion exchange water and dilute up to the mark, mix well. Calculate the normality of $K_2Cr_2O_7$ using the relation.

<u>Part B: Estimation of Iron</u> - Pipette out 25 cm^3 of the iron solution (of TMT bar) into a clean conical flask. Add a quarter test tube of conc. HCl and heat the solution to boil. Add stannous chloride to the hot solution drop wise till the yellow solution turns colorless. Add 2 more drops to ensure complete reduction. Cool and add 1 t.t of water and ½ test tube of mercuric chloride.

Place a number of drops of freshly prepared Potassium ferry cyanide indicator on a wax paper. Add a small quantity of $K_2Cr_2O_7$ from the burette to the conical flask containing iron solution and mix well. Remove a drop of solution from the conical flask and bring it in contact with a drop of the indicator on the wax paper. The color of the indicator turns blue. Repeat this operation after adding 1cm^3 more drop of the mixture incontact with fresh drop of the indicator. The indicator turns blue as long as the titration is incomplete. Continue the titration by adding the mixture increments of 1 cm^3 at a time and testing as above till a drop of the mixture fails to produce a blue color with the indicator dropon the wax paper (Clean the glass rod after every test). Repeat the titration by taking another 25cm^3 of TMT bar solution.

$$\underset{\text{Ferric chloride}}{2FeCl_3} + \underset{\text{Stannous chloride}}{SnCl_2} \longrightarrow \underset{\text{Ferrous chloride}}{2FeCl_2} + \underset{\text{Stannic chloride}}{SnCl_4}$$

$$\underset{\text{Stannous chloride}}{SnCl_2} + \underset{\text{Mercuric chloride}}{2HgCl_2} \longrightarrow \underset{\text{Mercurous chloride}}{Hg_2Cl_2} + \underset{\text{Stannic chloride}}{SnCl_4}$$

$$\underset{\text{Ferrous}}{Fe^{2+}} \longrightarrow \underset{\text{Ferric}}{Fe^{3+}} + e \times 6 \text{.....} (1)$$

$$\underset{\text{Dichromate}}{Cr_2O_7^{2-}} + 14H^+ + 6e^- \longrightarrow \underset{\text{Chromic}}{2Cr^{3+}} + 7H_2O \text{.....} (2)$$

Adding (1) & (2)

$$6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \longrightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$$

Result: Amount of iron in given TMT bar =_____%

ESTIMATION OF ACID MIXTURE BY CONDUCTOMETRIC SENSOR

OBSERVATION & CALCULATION

Sl. No	Vol. of NaOH (cm^3)	Conductance (ohm^{-1} cm^{-1})		Sl. No	Vol. of NaOH (cm^3)	Conductance (ohm^{-1} cm^{-1})
1	0.0			12	5.5	
2	0.5			13	6.0	
3	1.0			14	6.5	
4	1.5			15	7.0	
5	2.0			16	7.5	
6	2.5			17	8.0	
7	3.0			18	8.5	
8	3.5			19	9.0	
9	4.0			20	9.5	
10	4.5			21	10.0	
11	5.0					

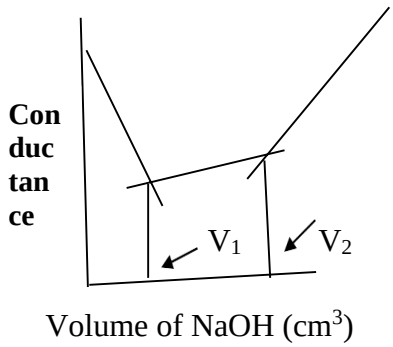
Normality of NaOH = 1.0 N

Volume of acid mixture taken = 50 cm^3

Volume of NaOH required to neutralize HCl = V_1 = ___cm^3

Volume of NaOH required to neutralize acid mixture V_2 = ___ cm^3

Volume of NaOH required to neutralize CH_3COOH =(V_2- V_1) =__cm^3



Normality of HCl = $\frac{N_{NaOH} \times V_1}{\text{Vol. of acid mixture}}$ = $\frac{_____ \times _____}{50\ cm^3}$ =_____N(a)

Weight of HCl present in 1dm^3 of the acid mixture = N_{HCl} x Eq. wt. of HCl

= a x 36.5 =_____g =--------- g

Normality of CH_3COOH = $\frac{N_{NaOH} \times (V_2 - V_1)}{\text{Volume of acid mixture}}$ = $\frac{_____}{50\ cm^3}$N (b)= N

Weight of CH_3COOH/dm^3 = $N_{CH3COOH}$ x Equivalent weight of CH_3COOH = bx60 =_____g

Result

Volume of NaOH required for neutralization of HCl = V_1 =_____cm^3

Volume of NaOH required for neutralization of CH_3COOH (V_2 - V_1) =_____cm^3

Date			
Experiment No.			

ESTIMATION OF ACID MIXTURE BY CONDUCTOMETRIC SENSOR

Aim: To estimate the amount of HCl and CH_3COOH in an acid mixture by conductometer sensor (conductometry)

Principle:

The neutralization point of acid solution in acid – base titration is determined by a change in the color of the indicator. Such a determination can be carried out by conductometric titration. In a conductometric titration sudden change in conductance of the solution near the neutralization point takes place. However the change is not sharp and hence the neutralization point is determined graphically by plotting conductivity against titre values. The principle underlying conductometric titration is the replacement of ions of a particular conductivity by ions of different conductivity during titration.

Conductometric titration may be applied for the determination of ions present in a mixture. In the titration of a mixture of a weak acid (CH_3COOH) and a strong acid (HCl) against a strong base (NaOH) the conductance initially decreases upon adding NaOH to acid mixture owing to the substitution of highly mobile H^+ ions (mobility: 350 $ohm^{-1}m^{-1}$) by the less mobile Na^+ ion (mobility: 50 $ohm^{-1}m^{-1}$) and this trend continues till all the H^+ ions of HCl are replaced i.e, the strong acid get neutralized. Later conductance slowly increases as CH_3COOH is converted into its salt, CH_3COONa. Further addition of NaOH raises the conductance steeply due to the presence of free OH^- ions (mobility; 198 $ohm^{-1}m^{-1}$).

Procedure :

Fill a micro burette with the standard NaOH solution. Pipette out 50 cm^3 of the given acid mixture into clean 100 cm^3 beaker. Place the conductivity sensor in the beaker, connect it to conductometer and measure the conductance. Add 0.5 cm^3 NaOH to beaker and measure conductance. With the addition of NaOH (with increments of 0.5 cm^3), initially conductance decreases and then rises slowly and finally rises sharply. Continue titration until the conductance is almost same as it was in the beginning of the titration. Plot a graph of conductance (Y- axis) versus volume of NaOH (X- axis). The point V_1 corresponds to neutralization of HCl and V_2 corresponds to neutralization of both hydrochloric and acetic acids. The difference in the volumes corresponding to the points V_2 and V_1 gives the volume required to neutralize CH_3COOH.

Result

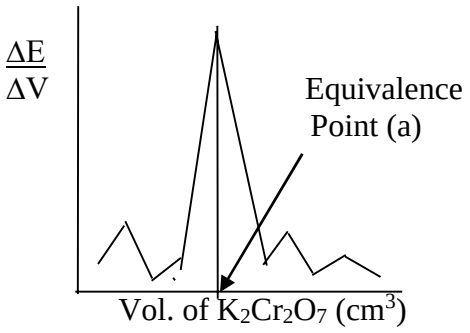
Volume of NaOH required for neutralization of HCl = V_1 =_____cm^3

Volume of NaOH required for neutralization of CH_3COOH (V_2 - V_1) =_____cm^3

ESTIMATION OF IRON IN RUST SAMPLE BY POTENTIOMETRIC SENSOR

OBSERVATION & CALCULATION

Volume of $K_2Cr_2O_7$ (cm^3)	EMF (mV)	ΔV	ΔE	$\frac{\Delta E}{\Delta V}$
0		NIL	NIL	NIL
0.5		0.5		
1.0		0.5		
1.5		0.5		
2.0		0.5		
2.5		0.5		
3.0		0.5		
3.5		0.5		
4.0		0.5		
4.5		0.5		
5.0		0.5		
5.5		0.5		
6.0		0.5		
6.5		0.5		
7.0		0.5		
7.5		0.5		
8.0		0.5		



From the graph volume of $K_2Cr_2O_7$ at equivalence point = -------cm^3 (a)

Normality of $K_2Cr_2O_7$ = 0.5 N (b)

Normality of FAS = $\frac{\text{Normality of } K_2Cr_2O_7 \text{ x Volume of } K_2Cr_2O_7 \text{ (from graph)}}{\text{Volume of FAS}}$

$= \frac{b \times a}{25}$ = _____ x _____ = _____ N (z)
25

Weight of FAS /dm^3 = Normality x gram equivalent weight of FAS

= z x 392 g = _____ x 392 = _____ g

Result: 1. From the graph, Volume of $K_2Cr_2O_7$ at equivalence point = _____ cm^3

2. Weight of FAS in given solution = _____ g

Date			
Experiment No.			

ESTIMATION OF IRON IN RUST SAMPLE BY POTENTIOMETRIC SENSOR

Aim: T o estimate the amount of iron in rust sample by potentiometric sensor (potentiometry)
Principle:

The estimation of amount of substance present in a solution by the measurement of emf is known as potentiometric titration**.** Here, emphasis is laid on the change in emf of cell as titrant added. A simple arrangement for potentiometric titration has a reference electrode (saturated calomel electrode) and an indicator electrode (platinum electrode).

The emf of the cell containing the solution is determined by adding increments of the titrant solution until the equivalence point reached and emf is determined after each addition. After reaching equivalence point, emf increases slightly on the excess addition of titrant.

For the redox potentiometric titration of FAS against $K_2Cr_2O_7$ the indicator electrode is a brightplatinum wire and the oxidizing agent ($K_2Cr_2O_7$) is placed in the burette. The reaction proceedsas follows.

$$2Cr_2O_7^{2-} + 14H^+ + Fe^{2+} \longrightarrow Fe^{3+} + Cr^{3+} + 7H_2O$$

Procedure:

Pipette out 25 cm^3 of the given FAS solution into a clean beaker and add 1 test tube of 2N H_2SO_4. Immerse platinum electrode and saturated calomel electrode assembly, connect to a pH meter/potentiometer in milli volts (mV) position and note down potential value. Fill a clean micro burette with the given standard $K_2Cr_2O_7$ solution.

Add $K_2Cr_2O_7$ solution with an increments of 0.5 cm^3 to FAS solution in the beaker. Stir the solution thoroughly and measure the emf after each addition. When the reaction reaches equivalence point, there is a maximum increase in the emf value. After the equivalence point, continue addition of $K_2Cr_2O_7$ and take 5 more readings.

Graph of $\Delta E/\Delta V$ against volume of $K_2Cr_2O_7$ is plotted. From the graph, volume of $K_2Cr_2O_7$ at equivalence point is determined.

Result: 1. From the graph, Volume of $K_2Cr_2O_7$ at equivalence point =_____cm^3

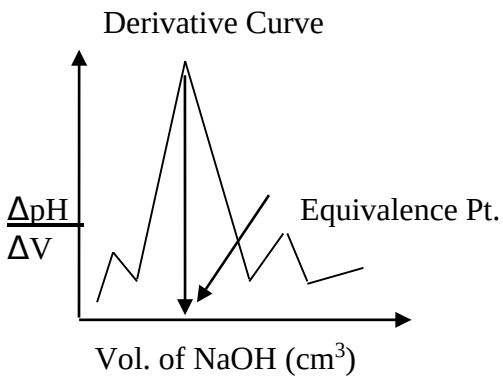
2. Weight of FAS in given solution =_____ g

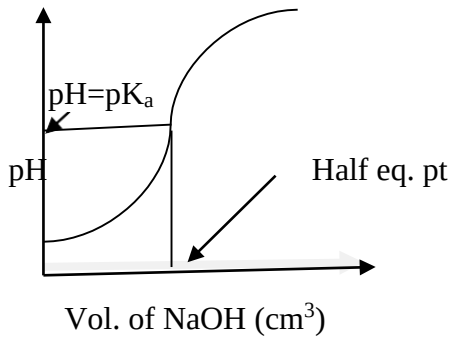
DETERMINATION OF pK_a OF VINEGAR USING pH SENSOR

OBSERVATION & CALCULATION:

Volume of NaOH (cm^3)	pH	ΔpH	ΔV	$\frac{\Delta pH}{\Delta V}$	Volume of NaOH (cm^3)	pH	ΔpH	ΔV	$\frac{\Delta pH}{\Delta V}$
0.0		NIL	NIL	NIL	5.5			0.5	
0.5			0.5		6.0			0.5	
1.0			0.5		6.5			0.5	
1.5			0.5		7.0			0.5	
2.0			0.5		7.5			0.5	
2.5			0.5		8.0			0.5	
3.0			0.5		8.5			0.5	
3.5			0.5		9.0			0.5	
4.0			0.5		9.5			0.5	
4.5			0.5		10.0			0.5	
5.0			0.5						

Nature of Graph:





Volume of NaOH at equivalence point =_____cm^3

Volume of NaOH at half equivalence point =_____cm^3

Result:

pH at half equivalence point = pKa of the given Vinegar solution = _____

Date			
Experiment No.			

DETERMINATION OF pK_a OF VINEGAR USING pH SENSOR

Aim: Determination of pKa of vinegar using pH sensor (glass electrode).

Principle:
During the titration of an acid with a base, the pH of the solution rises gradually at first and then more rapidly until at the equivalence point, there is a very sharp increase in pH for a very small quantity of added base. After the equivalence point, the pH increases only slightly on addition of excess base.

According to Henderson Hasselbalch equation,

$pH = pK_a + \log [salt] / [acid]$

Where Ka is dissociation constant of the acid and pKa = - log Ka. At half equivalence point, [salt] = [acid] and hence $pH = pK_a$. Therefore, pH at half equivalence point gives the pK_a of weak acid.

Procedure:

Pipette out 25 cm^3 of the given vinegar (Acetic acid) solution into a beaker. Immerse a pH sensor (glass and calomel electrode assembly) into the vinegar solution, connect it to a pH meter and measure pH of the solution. Blow the solution thoroughly using a glass tube and measure the pH after each addition of NaOH in the increments of 0.5 cm^3. Continue the addition of the base till there is only a slight increase in pH after equivalence point.

Plot a graph of $\Delta pH/\Delta V$ versus volume of NaOH added and find out equivalence point. Plot another graph of pH against volume of NaOH added and determine pH at half equivalence point. This is pKa vaule of given vinegar solution.

Result:

pH at half equivalence point = pKa of the given Vinegar solution = _____

ESTIMATION OF COPPER PRESENT IN E-WASTE BY OPTICAL SENSOR

OBSERVATION & CALCULATION

Flask No	Volume of NH_3 (cm^3)	Volume of effluent sample (cm^3)	Absorbance (Optical Density)
Blank (B)	5	0	
1	5	2.0	
2	5	4.0	
3	5	6.0	
4	5	8.0	
5	5	10.0	
TS	5	Test solution	

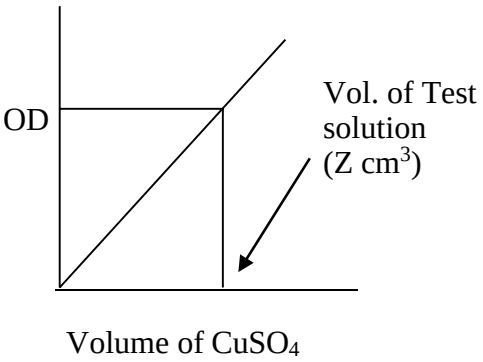
1 $cm^3$of copper sulphate solution contains 4 mg of $CuSO_4$

249.55 mg. of $CuSO_4$ = 63.54 mg. of Copper.

Therefore,

4 mg. of $CuSO_4$ $= \frac{63.54 \times 4}{249.55}$

= 1.018 mg. of Copper



i.e., 1 cm^3 of stock solution = 1.018 mg. of copper

Volume of test solution from the graph ('z') =_____ cm^3

Therefore, Weight of copper in the given test solution = 1.018 x 'Z' = ... mg

Result: Volume of test solution from the graph ('Z' cm^3) = ...cm^3

The concentration of the copper in the test solution = ...mg

Date			
Experiment No.			

ESTIMATION OF COPPER PRESENT IN E-WASTE BY OPTICAL SENSOR

Aim: To estimate the amount of Copper present in E-waste by Optical Sensor (Colorimetry)

Principle:

When a monochromatic light of intensity I_o is incident on a transparent medium, apart I_a of it is absorbed, I_r reflected and the remaining part I_t transmitted.

$$I_o = I_a + I_r + I_t$$

For a glass- air interface I_r is negligible.

Therefore, $I_o = I_a + I_t$

$I_t/I_o = T$ called the transmittance; $\log 1/T = \log I_o/I_t$ is called the absorbance or optical density A. The relation between absorbance A, concentration C (Expressed in mol/litre) and path length l (expressed in cm) is given by **Beer Lambert's law**.

$$A = \log I_o/I_t = €Ct$$

Where € is the molar extinction coefficient, t is the path length. C is the concentration of solution. If path length (t) is kept constant, then A œ C. Hence plot of absorbance against concentration gives a straight line.

A series of standard solutions of copper is treated with ammonia to get a blue **cuprammonium** complex and is diluted to a definite volume. The absorbance of each of these solutions is measured **at 620 nm**, since the complex shows maximum absorbance at this wavelength. The absorbance values are plotted against volume of copper sulphate (cm^3) to get a calibration curve.

A known volume of the test solution is treated with strong ammonia and diluted as above. The absorbance of this solution at 620 nm is measured and its concentration is determined from the calibration curve.

Procedure:

Transfer the given effluent sample (stock solution) to a burette and draw out 2, 4, 6, 8 and 10 cm^3 of the solution into 50 cm^3 volumetric flasks. Add 5 cm^3 of ammonia solution to each of them and dilute up to the mark with ion exchange water. Stopper the flasks and mix well. To the given test solution add 5 cm^3 ammonia, dilute up to the mark and mix well. Prepare a blank solution by diluting 5 cm^3 of ammonia solution in a 50 cm^3 volumetric flask up to the mark with ion exchange water and mix well.

Measure the absorbance of the solutions against blank at 620 nm using optical sensor. Draw a calibration curve by plotting optical density (OD) against volume of effluent sample (cm^3). Using the calibration curve find out the concentration of copper in the test solution and calculate the amount of copper.

Result: Volume of test solution from the graph ('Z' cm^3) =.....cm^3

DETERMINATION OF VISCOSITY CO-EFFICIENT OF CONDUCTIVE INKS

OBSERVATION & CALCULATION

	Trial I	Trial II	Trial III	Average
Lubricant sample (t_l) (in seconds)				
Water (t_w) (in seconds)				

Lab temperature =..... ^oC

η_w= Viscosity coefficient of water = ...milli poise

d_w = Density of water = ...g/cm^3

d_1 = Density of given lubricant sample =..... g/cm^3

Viscosity coefficient of given sample (η_l) $= \dfrac{t_l\, d_l \times \eta_w}{t_w\, d_w}$

=_____

=_____milli poise

Result:
Viscosity coefficient of the given lubricant sample = _____milli poise

Date			
Experiment No.			

DETERMINATION OF VISCOSITY CO-EFFICIENT OF CONDUCTIVE INKS

Aim: Determination of viscosity co-efficient of conductive inks (Ostwald's viscometer)

Principle:
A liquid flowing through cylindrical tube of uniform diameter is expected to move in the form of molecular layers. A layer close to surface is almost stationary while that at the axis of the tube moves faster than any other intermediate layer. A slow moving layer exerts a drag or friction on its nearest moving layer backwards. This property of a liquid by which it retards or opposes motion between layers is called viscosity.

The coefficient of viscosity is defined as the tangential force per unit area required to maintaina unit velocity gradient between any two successive layers of a liquid situated unit distance apart. The coefficient of viscosity of a liquid is given by poiscuille's formula,

$$\eta = \frac{\pi p r^4 t}{8vl}$$

Where v = volume of the liquid, r = radius of the tube, l = length of the tube, p is the pressure between the two ends of the tube, η is the coefficient of viscosity of the liquid. If equal volume of two different liquids are allowed to flow through the same tube with identical conditions then,

$$\frac{\eta_1}{\eta_w} = \frac{t_1 d_1}{t_w d_w}$$

The time t_1 taken by the given liquid to travel through a certain distance in the tube is determined. The time t_2 taken by a standard liquid (water) to travel through the same distance ismeasured. Knowing the densities d_1 and d_2 of the test liquid and the standard (water) and the viscosity coefficient of the standard, coefficient of viscosity of the test liquid can be calculated.

Procedure:
Take clean and dried viscometer and fix it vertically to a stand. Then transfer 10 cm^3 (using pipette) of given lubricant into the wider limb of the viscometer. Suck it above the upper mark ofthe viscometer and allow it to flow freely through the capillary. When the level of lubricant liquid just crosses the upper mark, start a stop clock and stop the stop clock when it just crosses the lower mark, and determine the time t_l taken in seconds by the lubricant to flow between fixed points. Repeat to get agreeing values for given lubricant. Pour it, wash with water, rinse with acetone and dry it. Fix the viscometer vertically to a stand and take 10 cm^3 of the water and suck it above the upper mark, allow it to flow freely through the capillary. When the level of water just crosses the upper mark, start a stop clock and stop it when the water just crosses the lower mark. Note the time of flow in seconds (t_w). Repeat to get agreeing values. By using t_l and t_w values determine the viscosity coefficient of given lubricant sample.

Result:
Viscosity coefficient of the given lubricant sample =_milli poise